

Chance-constrained optimal dispatch of integrated energy systems based on data-driven sparse polynomial chaos expansion[☆]

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ABSTRACT

In integrated energy systems, stochastic variations of different energy types of loads and the increasing penetration of renewable energy generation have resulted in considerable uncertainties. These uncertainties pose significant challenges to the economics and safe operation of integrated energy systems. Conventional deterministic methods of optimal dispatch overlook the effects of uncertainties, while stochastic optimization, though accounting for uncertainties, often yields conservative solutions that may adversely affect the economic operations. Chance-constrained optimization can effectively deal with uncertainties and provide more flexibility in balancing operational risks and benefits by expanding the feasible region. Hence, this paper proposes a chance-constrained optimal dispatch method for integrated energy systems and employs data-driven sparse polynomial chaos expansion method to enhance solving efficiency. First, the proposed chance-constrained optimization aims to ensure the optimal economic operation with an affordable security confidence level that balances safety and economics compared with deterministic and stochastic optimizations. The introduced data-driven sparse polynomial chaos expansion method enables the fast computing of the output response using only historical data, i.e., without knowledge pertaining to the distribution functions. Moreover, an improved iterative verification structure is proposed, which further improves the convergence speed and accuracy. Finally, the advantages and feasibility of the proposed method are verified using a test case and compared with those of deterministic and stochastic optimization. Results show that the proposed method successfully reduces the operational cost and controls violation probability.

Introduction

The global energy crisis and environmental problems caused by fossil fuels have prompted interest in clean, reliable, efficient, and economical energy technologies and infrastructure. To coordinate the construction and operation of energy systems, conventional separate energy networks for gas, electricity, and heat are interconnected via energy conversion equipment, thus forming integrated energy systems (IESs). An IES integrates the generation, distribution, storage, and utilization of diverse energy carriers, offering a versatile and effective energy provision for a wide range of consumer needs [1]. The synergistic benefits of IESs are significant, such as enhancing energy efficiency [2], facilitating the consumption of renewable resources [3], improving energy supply reliability and resilience [4], and contributing to environmental sustainability [5]. Therefore, IESs have been widely

recognized as a novel and promising paradigm for the development of energy systems in modern society.

The optimal dispatch of IESs is vital for enhancing their operational performance. However, coordinating the operations of different types of energy sources and equipment is challenging, and extensive studies have been conducted to address this issue. For small-scale IESs such as microgrids [6] and community-integrated energy systems [7], operation dispatch aims to optimally manage the energy balance by coordinating energy conversion devices [8], energy storage devices [9], and flexible loads [10]. For larger-scale IESs, such as regional IESs, the optimal dispatch is more complicated because of the topological constraints of the networks and the spatial distribution of multi-energy flows [11]. Novel methods have been developed to mitigate these problems, such as the optimal power flow calculation

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Nomenclature**Abbreviations**

AR	Absorption refrigerator
BCS	Bayesian compressive sensing
CCO	Chance-constrained optimization
CHP	Combined heat and power
DDSPCE	Data-driven sparse polynomial chaos expansion
DG	Distributed generation
EB	Electric boiler
ES	Electric storage
GB	Gas boiler
HS	Heat storage
IES	Integrated energy system
MCS	Monte-Carlo simulation
PCA	Principal component analysis
PCE	Polynomial chaos expansion
PDF	Probability density function
PV	Photovoltaic
SPCE	Sparse polynomial chaos expansion
SOC	State-of-charge
UEBM	Unified energy bus model

Indices

i, j	Indices of nodes
k	Indices of pipelines
t	Indices of time period

Variables

a_i	Expansion coefficients
C	Total operation cost of IES
$F_{b,i}^{\text{in}}$	i th energy flow entering energy bus b
$F_{b,j}^{\text{out}}$	j th energy flow leaving energy bus b
f	Flow rate vector of gas pipeline
H_k^c	Power consumption of the compressor k
i_{ij}	Current magnitude of branch ij
m	Flow rate vector
m_L	Injection flow rate vector
$m_{L,i}$	Injected flow rate of node i
m_k	Flow rate vector
P_i	Active power injection of bus i
P_{ij}	Active power flow of branch ij
p_i	Pressure of node i
Q_i	Reactive power injection of bus i
Q_{ij}	Reactive power flow of branch ij
T_i^s	Supply temperature of node i
T_i^o	Output temperature of node i
u_i	Voltage magnitude of bus i
W_i	Heat load power of node i
y^{ED}	Experimental design of the target response
$\Phi_i(x)$	Orthogonal polynomial basis

Parameters

A_h	Node-pipe matrix of heat network
A_g	Node-pipeline matrix of gas network
α	Relative specific heat capacity
B_h	Loop-pipe matrix of heat network
C_p	Specific heat capacity
C_k	Constant of the pipeline k
C_g	Gas price
C_e	Electricity price
$c_{b,i}$	Distribution coefficient of energy branch i on bus b
K	Resistance coefficient matrix of heat pipelines
K_c	Output coefficient of the compressor
L_k	Length of pipeline k
N	Experimental design size
R_k	Heat transfer coefficient of the pipeline k
R_{ij}	Resistance of branch ij
T_a	Atmospheric temperature
X_{ij}	Reactance of branch ij
Ω_b	Set of all electric branches
ϵ	Acceptable violation probability
$\eta_{e,ij}$	Efficiency of energy equipment between bus i and j

Functions

$\mathbb{P}[\cdot]$	Probability operator
$Q[\cdot]$	Quantile value

within IES, thereby hindering the applicability of the operation strategies obtained from deterministic optimization. For example, load and renewable energy fluctuations in electricity networks may result in voltage deviations [15] or power outages in the electrical equipment of other energy networks [16]. A gas-fired combined power and heat plant may require adjustment to its output to satisfy uncertain electricity and heat demands, which consequently affects the operational status of the gas network [17]. In these cases, deterministic optimization methods, which rely on fixed and known input parameters, cannot address complex uncertainties in IESs and may result in conservative, sub-optimal, or even infeasible operation strategies [18].

Significant efforts have been dedicated to address uncertainties in the dispatch of IESs, which can be classified into three categories: stochastic optimization [19], robust optimization [20], and interval optimization [21]. All of these methods have been widely utilized in the operation optimization of electric and integrated energy systems. Compared with stochastic optimization methods, robust and interval optimization methods do not require exact probabilistic distribution functions. Specifically, robust optimization methods address uncertainties by optimizing the worst case within a specified uncertainty set, whereas interval optimization methods require only the upper and lower bounds of random variables. However, compared with stochastic optimization, both robust and interval optimization methods are more conservative. Employing detailed probabilistic distribution functions that capture the characteristics of uncertain variables more precisely enables stochastic optimization methods to achieve higher accuracy and feasibility [22,23].

As a further developed stochastic optimization method, chance-constrained optimization (CCO) explicitly considers uncertainties by introducing probabilistic constraints, and further expands the feasible region by allowing the state variables to violate the constraints with a

of multi-energy networks [12], hierarchical operation optimization of large-scale IESs [13], and distributed optimization of multi-area IESs [14].

Nevertheless, a variety of energy types and the growing penetration of distributed energy resources have led to significant uncertainties

small predefined probability known as the confidence level [24]. Compared with stochastic and robust approaches, the chance-constrained optimization aims to find solutions that are less conservative and more in line with the specified risk levels [25]. Chance-constrained optimization allows the conservativeness of the optimal strategies to be adjusted, which provides better control over trade-offs between risk and benefits [26]. This can lead to more cost-effective and realistic solutions in practical applications.

The application of CCO to the optimal dispatch of IESs presents two major challenges. First, chance constraints in an IES present strong non-convexity and non-linearity because of the pipeline transmission characteristics of energy networks. A solution to this issue, as in power systems, is to transform chance constraints into closed-form deterministic or tractable problems that can be easily solved using linear or nonlinear programming solvers [27,28]. This transformation relies on deliberately designed modeling methods that are not applicable or are difficult to derive for IESs with diverse configurations and structures. Another solution involves using samples from chance constraints and converting them into multiple deterministic constraints, as represented by the Monte Carlo simulation (MCS) method [29]. However, sampling methods are generally computationally expensive, particularly for high-dimensional and complex IESs.

Second, as a type of stochastic optimization method, CCO requires knowledge pertaining to the probability density functions (PDFs) of random variables. However, various uncertainties arising from the loads and distributed generation in IESs cannot be depicted easily using exact probability distribution functions. Although some typical distributions can be adopted as approximations, such as the beta distribution for photovoltaic generation, Weibull distribution for wind generation, and uniform distribution for loads, these approximations may deviate from the actual uncertainty properties and thus result in significant errors. Moreover, complex correlations typically exist among uncertain factors, which further complicates the derivation of accurate PDFs [30].

This study is performed to devise an accurate and efficient CCO-based optimal dispatch method for electric-heat-gas IESs that considers arbitrary uncertainties from distributed generation and loads. In our previous study [31], we developed a probabilistic power-flow calculation method for IESs using sparse polynomial chaos expansion (SPCE), which is advantageous in terms of calculation efficiency and accuracy. In this study, the aforementioned method is further extended to account for arbitrary uncertainties using the concept of data-driven sparse polynomial chaos expansion (DDSPCE). The DDSPEC-based model is integrated into a chance-constrained optimal dispatch model of an IES to realize economic operation under uncertain scenarios with acceptable risks. A schematic illustration of the proposed method is shown in Fig. 1.

The main contributions of this study are summarized as follows:

(1) A probabilistic model of the IES operation state is developed based on data-driven sparse polynomial chaos expansion. The PDFs of the state variables of the IES can be efficiently obtained by considering the uncertainties from the distributed generation and loads with arbitrary distribution properties. The proposed model achieves an accuracy level similar to that of the Monte Carlo method, and the computational burden is reduced significantly.

(2) A day-ahead optimal dispatch method for the IES is proposed based on CCO. The probabilistic model of the IES is utilized to formulate chance constraints and evaluate the violation probability. The holistic CCO-based model is converted into a second-order cone-programming model to enable an easier solution. An improved iterative procedure is designed to ensure higher convergence precision.

(3) The day-ahead optimal dispatch of an electricity-gas-heat IES is realized using the proposed method, which balances between economic performance and operational safety. The conservation of the operation strategy is reduced compared with robust optimization, whereas the operational violation risk is controllable compared with conventional deterministic optimal dispatch methods.

Chance-constrained day-ahead optimal dispatch model of IES

This section introduces the optimization model of the chance-constrained day-ahead optimal dispatch.

Objective function

To account for the overall energy purchase cost of the IES, a combination of gas and electricity purchase costs is proposed as the objective function. Moreover, penalty terms are introduced in the objective function to restrict the second-order cone-relaxation model for electricity and gas systems, which is formulated as follows:

$$\min C = \sum_t \sum_i c_{g,t} f_{s,i,t} + \sum_t \sum_j c_{e,t} P_{j,t} + \lambda \left[\sum_t \sum_m i_{m,t} + \sum_t \sum_n (p_{n,t}^{\text{start}} - p_{n,t}^{\text{end}}) \right] \quad (1)$$

where C denotes the total operation cost of the IES; $c_{g,t}$ and $c_{e,t}$ denotes the gas and electricity price at period t , respectively; $f_{s,i,t}$ and $P_{j,t}$ denote the flow rate and power of the i th gas source and j th external grid, respectively; λ denotes the penalty term, which is an arbitrarily small positive value, i.e. 0.001; $i_{m,t}$ denotes the current of line m at period t ; and $p_{n,t}^{\text{start}}$ and $p_{n,t}^{\text{end}}$ denote the pressures of the start and end nodes of pipeline n at period t , respectively.

Constraints

This section introduces the optimal operation models for the IES, including the network constraints of electric, heat, and gas system, and the operation constraints of multi-energy station.

Electric network

(1) Electric flow balance constraints

A second-order cone programming model [32] is used to model the electric network, which is expressed as follows:

$$\sum_{ij \in \Omega_b} (P_{ij,t} - R_{ij} i_{ij,t}) + P_{j,t} = \sum_{jk \in \Omega_b} P_{jk,t} \quad (2)$$

$$\sum_{ij \in \Omega_b} (Q_{ij,t} - X_{ij} i_{ij,t}) + Q_{j,t} = \sum_{jk \in \Omega_b} Q_{jk,t} \quad (3)$$

$$u_{i,t} - u_{j,t} + (R_{ij}^2 + X_{ij}^2) i_{ij,t} = 2 (R_{ij} P_{ij,t} + X_{ij} Q_{ij,t}) \quad (4)$$

$$\left\| \begin{matrix} 2P_{ij,t} \\ 2Q_{ij,t} \end{matrix} \right\| \leq i_{ij,t} + u_{i,t} \quad (5)$$

$$P_{i,t} = P_{i,t}^{\text{DG}} - P_{i,t}^{\text{L}} \quad (6)$$

$$Q_{i,t} = Q_{i,t}^{\text{DG}} - Q_{i,t}^{\text{L}} \quad (7)$$

where Ω_b denotes the set of all branches; $i_{ij,t}$ denotes the current magnitude of branch ij ; $u_{i,t}$ denotes the voltage magnitude of bus i ; $P_{ij,t}$ and $Q_{ij,t}$ denote the active and reactive power flows of branch ij , respectively; $P_{j,t}$ and $Q_{j,t}$ denote the total active and reactive power injection at bus j , respectively; and R_{ij} and X_{ij} denote the resistance and reactance of branch ij , respectively.

Constraints (2) and (3) represent the power balance of node i in period t . Constraint (4) expresses Ohm's law for branch ij at period t , where $u_{i,t}$ denotes the voltage magnitude at node i . The current magnitude of each line can be determined using (5), which is further relaxed to a second-order cone constraint. Constraints (6) and (7) indicate the total active and reactive power injection of node j at period t , respectively.

(2) Electric system operation constraints

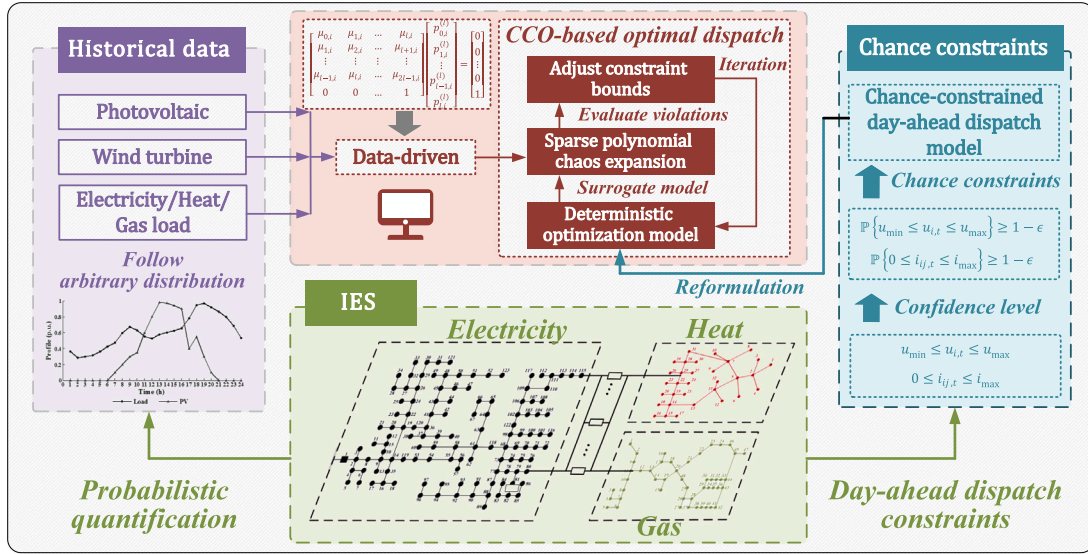


Fig. 1. Schematic illustration of the proposed chance-constrained optimal dispatch method of IES.

The system voltage should be maintained within an acceptable range for the optimal dispatch of the IES, which is formulated as:

$$u_{\min} \leq u_{i,t} \leq u_{\max} \quad (8)$$

where $u_{\max}$ and $u_{\min}$ denote the upper and lower limits of the nodal voltage, respectively.

The maximum permissible line current constraint is expressed as:

$$0 \leq i_{ij,t} \leq i_{ij,\max} \quad (9)$$

where $i_{ij,\max}$ denotes the upper current limit of branch ij .

Constraints (8) and (9) limit the violation of state variables in the electric subsystem of the IES; however, not all constraints are strictly unacceptable in actual industrial applications. Therefore, chance constraints can be applied to (8) and (9) to allow the operators to obtain additional scenarios in the optimal dispatch to balance between operational cost and security, which can be written as:

$$\mathbb{P}\{u_{\min} \leq u_{i,t} \leq u_{\max}\} \geq 1 - \epsilon \quad (10)$$

$$\mathbb{P}\{0 \leq i_{ij,t} \leq i_{\max}\} \geq 1 - \epsilon \quad (11)$$

where $\mathbb{P}$ denotes the probability operator; ϵ denotes the acceptable violation probability, which is typically set to be a small number (e.g., 1%, 5%, and 10%) to ensure a secure operation of the systems, and $1 - \epsilon$ is called the confidence level in chance constraints.

(3) Operation constraints of distributed generator (DG) and electric storage (ES)

The operational constraints of DGs and ES are considered in this study using the typical operation dispatch model [33].

In actual industrial applications, the operational constraints in an IES can be classified into soft and hard constraints [27]. For soft constraints, such as voltage and current magnitude constraints, as shown in (8) and (9), respectively, a constraint violation implies an actual under- or over-voltage, or a transmission line overload. Generally, a low violation probability of the soft constraints is acceptable in actual applications. However, hard constraints such as storage and generation constraints, are physically impossible to violate.

Therefore, because the operational constraints of DGs and ES are hard constraints that do not contain chance constraints, the model description of DGs and ES is introduced in Appendix “Operation constraints of DG, ES, and HS”.

Heat network

(1) Heat flow balance constraints

The constraints of a heat network are typically a set of nonlinear functions composed of hydraulic and thermal models [34].

$$A_h m - m_L = 0 \quad (12)$$

$$B_h K m |m| = 0 \quad (13)$$

$$W_{i,t} = C_p m_{L,i,t} (T_{i,t}^s - T_{i,t}^o) \quad (14)$$

$$T_{k,t}^{\text{end}} = (T_{k,t}^{\text{start}} - T_a) \exp(-R_k L_k / C_p m_{k,t}) + T_a \quad (15)$$

$$(\sum m_{\text{out},k}) T_{\text{start}} = \sum (m_{\text{in},k} T_{\text{end}}) \quad (16)$$

where A_h denotes the node-pipe connection matrix of the heat network; m denotes the flow rate vector; m_L denotes the injection flow rate vector; B_h denotes the loop-pipe connection matrix; K denotes the resistance coefficient matrix; $W_{i,t}$ denotes the heat load power of node i at period t ; C_p denotes the specific heat capacity of water; $m_{L,i,t}$ denotes the load flow rate of node i at period t ; $T_{i,t}^s$ and $T_{i,t}^o$ denote the supply and output temperatures of node i at period t , respectively; $T_{k,t}^{\text{start}}$ and $T_{k,t}^{\text{end}}$ denote the start and end temperatures of pipeline k at period t , respectively; T_a denotes the atmospheric temperature; R_k denotes the heat transfer coefficient of the pipeline k ; L_k denotes the length of pipeline k , and $m_{k,t}$ denotes the flow rate within pipeline k at period t ; and $m_{\text{in},k}$ and $m_{\text{out},k}$ denote the rate at which flow enters and departs from the node of pipeline k , respectively.

Constraints (12) and (13) show the hydraulic constraints of the heat network, and constraints (14)–(16) express the thermal model that describes the thermal behavior of the loads, pipelines, and the thermal balance of the heat system.

In many studies, a constant mass flow rate and variable temperature control method are typically used, which does not significantly affect the calculation of thermal losses in the heat network and has been demonstrated to be effective in industrial applications [35]. Thus, constraints (14)–(16) are regarded as linear constraints in this study.

(2) Heat system operation constraints

The secure constraints of the supply and return temperatures are constructed in constraints (17) and (18), respectively.

$$T_{\min}^s \leq T_{i,t}^s \leq T_{\max}^s \quad (17)$$

$$T_{\min}^r \leq T_{i,t}^r \leq T_{\max}^r \quad (18)$$

where $T_{\max}^s$ and $T_{\min}^s$ denote the maximum and minimum limits of the supply temperature, respectively; and $T_{\max}^r$ and $T_{\min}^r$ denote the maximum and minimum limits of the return temperature, respectively.

Constraints (17) and (18) can be converted into chance constraints as follows:

$$\mathbb{P} \left\{ T_{\min}^s \leq T_{i,t}^s \leq T_{\max}^s \right\} \geq 1 - \epsilon \quad (19)$$

$$\mathbb{P} \left\{ T_{\min}^r \leq T_{i,t}^r \leq T_{\max}^r \right\} \geq 1 - \epsilon \quad (20)$$

(3) Operational constraints of heat storage (HS)

The operational constraints of HS are considered in this study. Similar to the case of DGs and ES, the operational constraints of HS are hard constraints, in which no chance constraints are considered. The operational model of the HS used in this study is presented in Appendix “Operation constraints of DG, ES, and HS”.

Gas network

(1) Gas flow balance constraints

The gas network constraints include of the network constraints of gas pipelines and the operational constraints of gas compressors. The network constraints of gas pipeline under steady-state [36] can be expressed as:

$$\mathbf{A}_g \mathbf{f}_t = \mathbf{0} \quad (21)$$

$$f_{k,t} = C_k \sqrt{(p_{i,t})^2 - (p_{j,t})^2} \quad (22)$$

$$H_{k,t}^{\text{COM}} = K_c \left[\left(\frac{p_{j,t}}{p_{i,t}} \right)^{\frac{\alpha-1}{\alpha}} - 1 \right] \quad (23)$$

where $\mathbf{A}_g$ denotes the node-pipe connection matrix of the gas network, $\mathbf{f}_t$ denotes the flow rate vector of all gas pipelines at period t ; C_k denotes a constant of the pipeline k , which comprise the temperature, length, diameter, roughness, etc.; $p_{i,t}$ and $p_{j,t}$ denote the pressure of nodes i and j at period t , respectively; K_c denotes the output coefficient of the compressor; $H_{k,t}^{\text{COM}}$ denotes the power consumption of compressor k at period t ; α denotes the relative specific heat capacity; and $R_{k,\max}$ and $R_{k,\min}$ denote the maximum and minimum compression ratios of the compressor, respectively.

Constraint (21) represents the flow-rate balance of the gas network. Constraint (22) represents the pressure drop model of a gas pipeline. Constraint (23) describes the power consumption model of the compressors.

However, constraint (23) cannot easily yield solution for the gas flow because of the non-convex constraint. Thus, constraint (23) should be relaxed as a second-order cone constraint [37] expressed as:

$$\left\| \frac{f_{k,t}}{C_k} \right\| \leq p_{i,t} \quad (24)$$

(2) Gas system operation constraints

The secure constraints of the nodal pressure, flow rate, and compression ratio are constructed in constraints (25)–(27).

$$p_{\min} \leq p_{i,t} \leq p_{\max} \quad (25)$$

$$0 \leq f_{k,t} \leq f_{\max} \quad (26)$$

$$R_{k,\min} \leq \frac{p_{j,t}}{p_{i,t}} \leq R_{k,\max} \quad (27)$$

where $p_{\max}$ and $p_{\min}$ denote the maximum and minimum limits of the nodal pressure, respectively; $f_{\max}$ denotes the maximum flow rate of the pipeline; and $R_{k,\max}$ and $R_{k,\min}$ denote the maximum and minimum compression ratios of the compressor, respectively. Constraints (25) and (26) can be converted into chance constraints as follows:

$$\mathbb{P} \{ p_{\min} \leq p_{i,t} \leq p_{\max} \} \geq 1 - \epsilon \quad (28)$$

$$\mathbb{P} \{ 0 \leq f_{k,t} \leq f_{\max} \} \geq 1 - \epsilon \quad (29)$$

Energy station

(1) Energy station balance constraints

The unified energy bus model (UEBM) proposed in our previous study [38] is used to describe the energy station. An example of the unified energy bus model is shown in Fig. 2, where CHP denotes combined heat and power, GB denotes gas boiler, and AR denotes absorption refrigerator. A typical energy station in an IES can be formulated using the energy bus balance function (30), equipment efficiency function (31), and distribution coefficient function (32), thus resulting in a matrix-based mathematical model of the energy station.

$$\sum_{i=1}^m F_{b,i,t}^{\text{in}} = \sum_{j=1}^m F_{b,j,t}^{\text{out}} \quad (30)$$

$$F_{e,i,t}^{\text{in}} \eta_{e,i,j,t} = F_{e,j,t}^{\text{out}} \quad (31)$$

$$F_{b,i,t}^{\text{in}} = c_{b,i,t} L_{b,t} \quad (32)$$

where $F_{b,i,t}^{\text{in}}$ denotes the i th energy flow entering energy bus b at period t ; $F_{b,j,t}^{\text{out}}$ denotes the j th energy flow departing from energy bus b at period t ; m and n denote the numbers of branches that enter and depart from energy bus b ; $F_{e,i,t}^{\text{in}}$ denotes the i th energy flow entering equipment e at period t ; $F_{e,j,t}^{\text{out}}$ denotes the j th energy flow leaving equipment e at period t ; $\eta_{e,i,j,t}$ denotes the efficiency of energy equipment connected between bus i and j ; and $c_{b,i,t}$ denotes the distribution coefficient of energy type i on bus b at period t .

(2) Energy station operation constraints

The secure constraints of the input and output power of the energy equipment can be expressed as:

$$F_{e,\min}^{\text{in}} \leq F_{e,t}^{\text{in}} \leq F_{e,\max}^{\text{in}} \quad (33)$$

$$F_{e,\min}^{\text{out}} \leq F_{e,t}^{\text{out}} \leq F_{e,\max}^{\text{out}} \quad (34)$$

where $F_{e,\max}^{\text{in}}$ and $F_{e,\min}^{\text{in}}$ denote the minimum and maximum input powers of the energy equipment e , respectively; and $F_{e,\max}^{\text{out}}$ and $F_{e,\min}^{\text{out}}$ denote the minimum and maximum output powers of energy equipment e , respectively.

Therefore, the chance-constrained day-ahead optimal dispatch model is formulated as follows:

$$\begin{aligned} \text{obj. } & (1) \\ \text{s.t. } & (2)–(7), (10)–(16), (19)–(21), (23), (24), (27)–(34) \end{aligned} \quad (35)$$

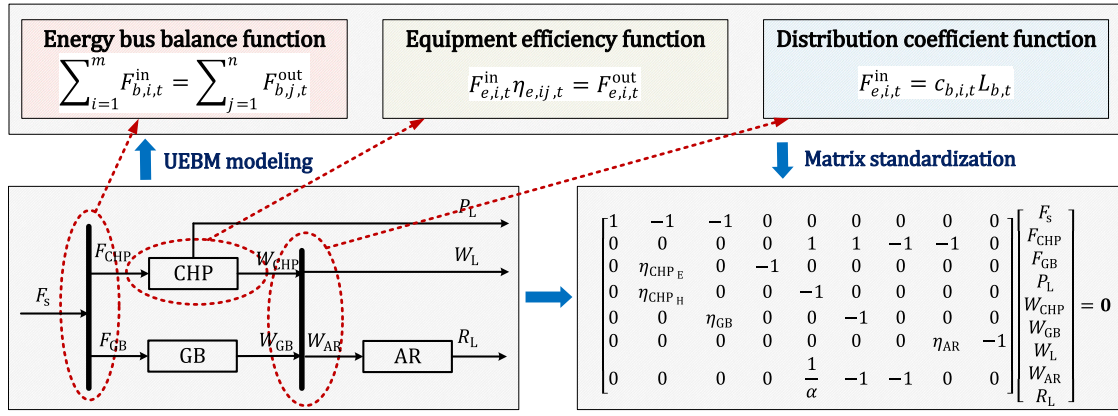


Fig. 2. Structure of the unified energy bus model.

Data-driven probabilistic modeling of IES operation state

Sparse polynomial chaos expansion

According to polynomial chaos expansion (PCE) theory [39], the chance-constrained day-ahead optimal dispatch model (35) can be regarded as a black box model, and the function of its response y can be approximated by a surrogate model using a multidimensional PCE function $P(x)$ as follows:

$$y = f(x) \approx P(x) = \sum_i a_i \Phi_i(x) \quad (36)$$

where $x = [x_1, x_2, \dots, x_n]$ denotes an n -dimensional independent random vector; $\Phi_i(x)$ denotes the orthogonal polynomial basis of the response function, which is determined by the stochastic characteristics of the input random variable x_i ; and a_i denotes the unknown expansion coefficients.

The polynomial basis of a single random variable can be selected directly based on its distribution. The typical distributions of the random variables and their polynomial bases corresponding to the distributions has been summarized as the Wiener-Askey scheme [40]. Subsequently, the polynomial bases of each independent random variables can be decided, which is expressed as:

$$\phi = \{\phi_i^j(x_i)\} \quad i = 1, 2, \dots, n; j = 1, 2, \dots, r \quad (37)$$

Here, note that ϕ is the basis of the independent random variables, which should be distinguished from the bases of the output response Φ . Thus, all the n random variables will feature $r \times n$ bases. Subsequently, the i th polynomial basis $\phi_i(x)$ of the response function can then be calculated as:

$$\Phi_i(x) = \prod_{i=1}^n \phi_i^{(k_i)}(x_i) \quad (38)$$

where k_i satisfies $0 \leq \sum_{i=1}^n k_n \leq r$.

Subsequently, we can compute the expansion coefficient a_i to derive a surrogate model using the sampling data. The sampling process, often referred to experimental design, involves sampling random input variables and performing calculations for each scenario. Consequently, the problem can be reformulated as the following residual minimization problem:

$$\hat{a} = \argmin \sum_{i=1}^N [y^{ED}(x^{(i)}) - a^T \Phi(x^{(i)})]^2 \quad (39)$$

where N denotes the sampling size of experimental design; $x^{(i)}$ denotes the i th sampling dataset; and y^{ED} denotes the experimental design of the objective function Y of the day-ahead optimal dispatch. Thus, The polynomial expression of the objective function could finally be obtained after the coefficients are determined. In the conventional

SPCE, the orthogonal polynomial basis is selected based on the probability distribution of the random variable. However, datasets, instead of probabilistic distributions of random inputs (e.g. wind speed), may be easier to acquire from IESs. Therefore, we use the data-driven SPCE method [41] which constructs a polynomial basis from data instead of from a random distribution, thus effectively extending the limitation of PCE to arbitrary probability distributions.

Data-driven sparse polynomial chaos expansion

For a one-dimensional random variable ξ , suppose that $Q^{(k)}(\xi)$ represents its orthogonal polynomials, which can be written as:

$$Q^{(k)}(\xi) = \sum_{i=0}^k q_i^{(k)} \xi^i \quad (40)$$

where $q_i^{(k)}$ is the expansion coefficient of $Q^{(k)}(\xi)$.

The corresponding orthogonal polynomial basis is constructed using the moments of each input variable. In the case of a monic polynomial, based on its orthogonal conditions, (41) and (42) are expressed as follows:

$$\begin{cases} \int_{\xi \in \Omega} Q^{(k)}(\xi) Q^{(l)}(\xi) \mu(d\xi) = \delta_{kl} & \forall k, l = \overline{0, d} \\ \int_{\xi \in \Omega} Q^{(k)}(\xi) Q^{(l)}(\xi) \mu(d\xi) = 0 & \forall k \neq l \end{cases} \quad (41)$$

$$q_k^{(k)} = 1 \quad \forall k \quad (42)$$

According to (41) and (42), $q_0^{(0)} = 1$ can be directly obtained, followed by $Q^{(0)}(\xi) = q_0^{(0)} = 1$. Based on the orthogonality between the zeroth- and first-degree polynomials, the following equation can be obtained:

$$\int_{\xi \in \Omega} Q_0^{(0)} \left[\sum_{i=0}^1 q_i^{(1)} \xi^i \right] \mu(d\xi) = 0 \quad q_1^{(1)} = 1 \quad (43)$$

According to the orthogonality of any k -order polynomial and all its lower-order polynomials, (44) can be obtained by continuing the aboveground push process.

$$\begin{aligned} \int_{\xi \in \Omega} Q_0^{(0)} \left[\sum_{i=0}^k q_i^{(k)} \xi^i \right] \mu(d\xi) &= 0 \\ \int_{\xi \in \Omega} \left[\sum_{i=0}^1 q_i^{(1)} \xi^i \right] \left[\sum_{i=0}^k q_i^{(k)} \xi^i \right] \mu(d\xi) &= 0 \\ &\vdots \\ \int_{\xi \in \Omega} \left[\sum_{i=0}^{k-1} q_i^{(k-1)} \xi^i \right] \left[\sum_{i=0}^k q_i^{(k)} \xi^i \right] \mu(d\xi) &= 0 \\ q_k^{(k)} &= 1 \end{aligned} \quad (44)$$

The definition of k -order orthogonal polynomial in (44) involves all the low-order polynomials $Q^{(k-1)}$, $Q^{(1)}$, and $Q^{(0)}$. Thus, (44) can be simplified to (45) by substituting the first expression into the second expression, the first and second expressions into the third expression, etc.

$$\begin{aligned} \int_{\xi \in \Omega} \left[\sum_{i=0}^k q_i^{(k)} \xi^i \right] \mu(d\xi) &= 0 \\ \int_{\xi \in \Omega} \left[\sum_{i=0}^k q_i^{(k)} \xi^{i+1} \right] \mu(d\xi) &= 0 \\ &\vdots \\ \int_{\xi \in \Omega} \left[\sum_{i=0}^k q_i^{(k)} \xi^{i+k-1} \right] \mu(d\xi) &= 0 \\ q_k^{(k)} &= 1 \end{aligned} \quad (45)$$

Therefore, the k th moment of the input variable μ_k can be defined as follows:

$$\mu_k = \int_{\xi \in \Omega} \xi^k \mu(d\xi), k = 0, 1, 2, \dots, 2d-1 \quad (46)$$

Using (46), (45) can be reformulated into the following equation:

$$\begin{aligned} \sum_{i=0}^k q_i^{(k)} \mu_i &= 0, \sum_{i=0}^k q_i^{(k)} \mu_{i+1} = 0 \quad \vdots \\ \sum_{i=0}^k q_i^{(k)} \mu_{i+k-1} &= 0, \\ q_k^{(k)} &= 1 \end{aligned} \quad (47)$$

Subsequently, (47) can be written in matrix form as follows:

$$\begin{bmatrix} \mu_0 & \mu_1 & \dots & \mu_k \\ \mu_1 & \mu_2 & \dots & \mu_{k+1} \\ \vdots & \vdots & \vdots & \vdots \\ \mu_{k-1} & \mu_k & \dots & \mu_{2d-1} \\ 0 & 0 & \dots & 1 \end{bmatrix} \begin{bmatrix} q_0^{(k)} \\ q_1^{(k)} \\ \vdots \\ q_{d-1,i}^{(k)} \\ q_{d,i}^{(k)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \quad (48)$$

where the k th moment of the random input $\mu_k, k = [0, 1, \dots, 2d-1]$ can be approximated from the samples based on $\mu_k = \frac{1}{M_p} \sum_{m=1}^{M_p} \zeta_{m,i}^k$, M_p denotes the sample size, and $\zeta_{m,i}$ are the sample points of an arbitrary input random variable ζ_i . Subsequently, using the raw dataset or distributions of ζ_i , the raw moment of ζ_i can be obtained. It can be seen from (48) that the availability of the k th order polynomial expansion requires the calculation of moments $\mu_0, \mu_1, \dots, \mu_{2d-1}$. ($k = 0, 1, 2, \dots, d$). Thus, to construct a d th order SPCE model, the moment up to $(2d-1)$ th order of the raw data needs to be calculated.

Calculation of expansion coefficients

According to (39), once the polynomial basis has been determined, the expansion coefficient a_i can be calculated to obtain the polynomial surrogate model based on the sampling data. In our previous study [31], we proposed a Bayesian compressive sensing (BCS) method to realize fast and accurate calculations of the SPCE model. In this study, the BCS was applied to solve for the expansion coefficient $\hat{a}$ of the surrogate model obtained using the DDSPCE method.

For a given polynomial basis and initial sampling dataset, the expression of the likelihood function is as follows:

$$P(\mathbf{Y}|\Phi, \mathbf{a}, \mu) = N(\mathbf{Y}|\mathbf{a}\Phi, \mu^{-1}) \quad (49)$$

The problem stated in (49) is analogous to imposing a Laplace prior on the coefficient $\mathbf{a}$. Therefore, hierarchical priors are employed to tackle this problem [42].

$$P(\mathbf{a}|\lambda) = \prod_{i=1}^P N(c_i|0, \lambda_i) \quad (50)$$

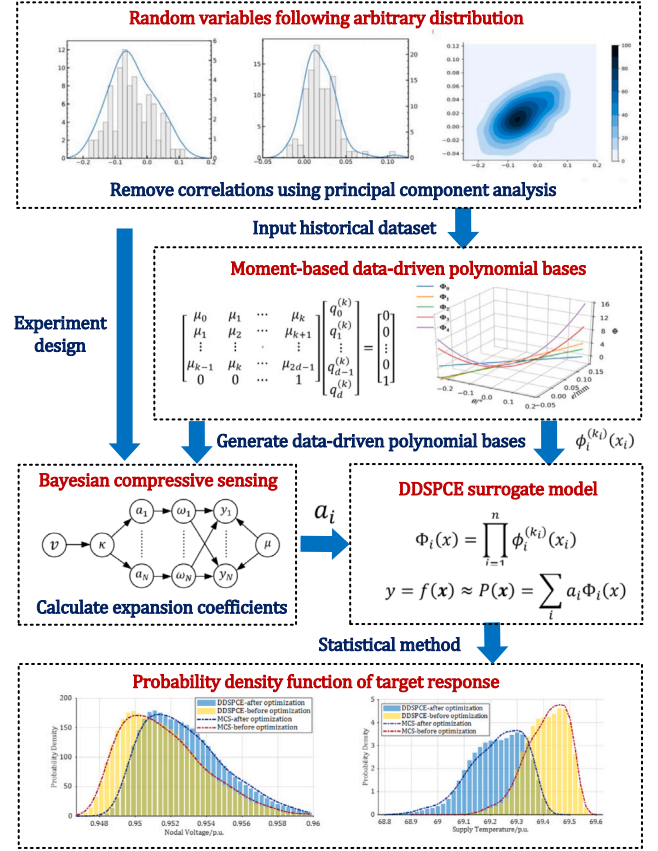


Fig. 3. Procedure for data-driven probabilistic modeling of IES operation state.

$$P(\lambda_i|\kappa) = \Gamma(\lambda_i|1, \frac{\kappa}{2}) = \frac{\kappa}{2} \exp(-\frac{\kappa \lambda_i}{2}) \quad (51)$$

Thus, the posterior distribution of all parameters can be obtained by using the Bayesian inference:

$$P(\mathbf{a}, \lambda, \kappa, \mu|\mathbf{y}) = \frac{P(\mathbf{y}|\mathbf{a}, \mu)P(\mathbf{a}|\lambda)P(\lambda|\kappa)P(\mu)\kappa}{P(\mathbf{y})} \quad (52)$$

Thus, the posterior distribution of $\mathbf{a}$ can be obtained from $P(\mathbf{a}, \lambda, \kappa, \mu|\mathbf{y})$. It follows a Gaussian distribution $N(\mathbf{a}|\mu_a, \Sigma_a)$. Here,

$$\delta_a = \Sigma_c \mu \Phi^T \mathbf{y} \quad (53)$$

$$\Sigma_a = (\mu \Phi^T \Phi + \text{diag}(\frac{1}{\lambda_i}))^{-1} \quad (54)$$

Therefore, the hyper parameters in (53) and (54) can be calculated by maximizing $P(\mathbf{y}, \lambda, \kappa, \mu)$. A fast Laplace algorithm is applied to reduce the computational cost by updating a single λ_i rather than total λ in each step. Thus, the update of δ_c and Σ_c could become more efficient. Further details regarding the algorithm are available in [31].

Combining all the methods introduced in Section “Data-driven probabilistic modeling of IES operation state”, Fig. 3 presents an overall flowchart of the proposed data-driven probabilistic modeling of the IES operation state. First, based on the raw historical data of the random input variables, a decorrelated dataset of random variables was constructed using the principal component analysis (PCA) algorithm. Subsequently, these datasets were used to generate moment-based polynomial bases for the surrogate models. Next, a series of input data samples was selected as the experimental design for the BCS to calculate the expansion coefficients. A surrogate model for the selected response can be established by combining the coefficient with a polynomial basis. Finally, a probability density function can be obtained by applying statistical methods to the surrogate model.

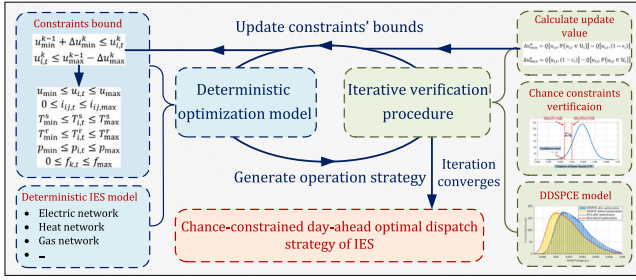


Fig. 4. Iterative verification procedure for chance-constrained day-ahead optimal dispatch.

Chance-constrained day-ahead optimal dispatch based on DDSCPE

As the chance-constrained optimal dispatch model and the DDSPCE have been introduced in the previous sections, the scheme of chance-constrained day-ahead optimal dispatch based on DDSCPE will be introduced in this section.

Solving scheme of chance-constrained day-ahead optimal dispatch model

The proposed optimization model contains chance constraints that cannot be addressed directly using deterministic optimization solvers. In existing studies, one-shot [43] and iterative [44] methods are typically used to manage chance constraints. The one-shot method reformulates chance constraints into deterministic constraints and solves the problem directly, which is a more suitable structure for intrusive methods (e.g. generalized polynomial chaos). Meanwhile, the iterative method is typically used in non-intrusive methods such as the MCS. The core idea of the iterative method is to solve the deterministic problem without chance constraints multiple times while verifying the satisfaction of chance constraints in each iteration and adjusting the boundaries of the deterministic problem for the next iterative step. When the iterations converge, the solution to the deterministic problem, which is also the solution to the chance-constrained problem, is obtained.

In this study, the DDSPCE method was used in conjunction with an iterative computational framework. Owing to its rapid computation of probability density functions, DDSPCE can significantly facilitate the verification speed of chance constraints and effectively solve chance-constrained problems. The iterative solution framework for the chance-constrained optimization problem based on DDSPCE is illustrated in Fig. 4.

Iterative verification algorithm of chance constraints

The conventional iterative method assumes that the algorithm converges when all chance constraints are satisfied, without guaranteeing that the maximum probability of violation in the system converges at a specified confidence level, which may result in a relatively conservative operating strategy. In this section, an iterative algorithm with an improved convergence condition and an iterative format that can guarantee better convergence at the confidence level are described.

Using constraint (8) as an example, in each iterative step, its chance constraint (10) is regarded as a deterministic constraint, which is represented as:

$$u_{\min}^{k-1} + \Delta u_{\min}^k \leq u_{i,t}^k \quad (55)$$

$$u_{i,t}^k \leq u_{\max}^{k-1} - \Delta u_{\max}^k \quad (56)$$

where $u_{\min}^{k-1}$ and $u_{\max}^{k-1}$ denote the lower and upper bounds of the constraint in step $k-1$, respectively; and $\Delta u_{\min}^k$ and $\Delta u_{\max}^k$ denote the update

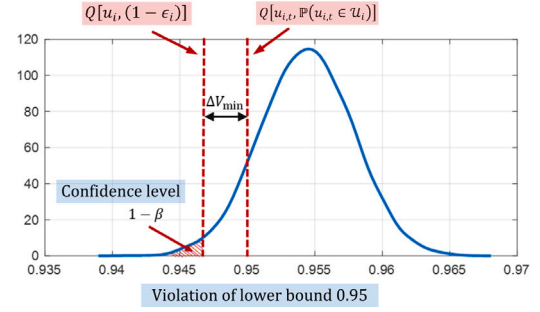


Fig. 5. Example of quantile values under violation of lower bound.

value of the lower and upper bounds in step k , respectively, which can be calculated as:

$$\Delta u_{\min}^k = Q[u_{i,t}, P(u_{i,t} \in U_i)] - Q[u_{i,t}, (1 - \epsilon_i)] \quad (57)$$

$$\Delta u_{\max}^k = Q[u_{i,t}, (1 - \epsilon_i)] - Q[u_{i,t}, P(u_{i,t} \in U_i)] \quad (58)$$

where $[u_{i,t}, P(\cdot)]$ denotes the quantile value for the distribution of $u_{i,t}$ at a probability value of P . An example of quantile values violating the lower bound is shown in Fig. 5. A detailed iterative algorithm is presented in Algorithm 1.

Algorithm 1 Iterative solution algorithm

- 1: Initialization: Set all of the difference values $\Delta x_{i,\min} = \Delta x_{i,\max} = 0$, and iteration count $k = 1$, and the convergence precision δ .
- 2: Solve the deterministic optimal dispatch problem defined with bounds $x_{i,\min}$ and $x_{i,\max}$ and obtain the optimal operation status x_k of the system.
- 3: Use the probability distribution of the state variables x_k to verify whether the chance constraints are satisfied.
- 4: Calculate $d = \max\{|\mathbb{P}\{x_{i,t} \leq x_{i,\min} \& x_{i,t} \geq x_{i,\max}\} - \epsilon|, i = 1, \dots, n$.
- 5: If $d > \delta$, increase iterative count $k = k + 1$, calculate $\Delta x_{i,\min}^k$ or $\Delta x_{i,\max}^k$ using (57) and (58), respectively, and update the bound $x_{i,\min}^k$ or $x_{i,\max}^k$ using the deterministic calculation result $x_{i,\det}^{k-1}$ instead of $x_{i,\min}^{k-1}$ or $x_{i,\max}^{k-1}$, represented as $x_{i,\min}^k = x_{i,\det}^{k-1} + \Delta x_{i,\min}^k$ and $x_{i,\max}^k = x_{i,\det}^{k-1} + \Delta x_{i,\max}^k$. Proceed to Step 2.
- 6: If $d < \delta$, algorithm converges.

Overall procedure of DDSPCE for chance-constrained optimal dispatch of IES

The structure of the DDSPCE proposed herein is shown in Fig. 6, and the detailed steps are summarized as follows:

Step (1): Input the deterministic network parameters of the IES, including the topology and parameters of pipes or lines in energy sub-systems, and the efficiency parameters of energy stations. Additionally, input the history data of the random variables considered in the IES, including the PV, wind turbines, and different types of loads.

Step (2): Calculate the moments of the historical dataset and generate data-driven polynomial bases for random variables. The bases are calculated only once at the beginning and need not be updated during the iterative process.

Step (3): Establish a deterministic model for the day-ahead dispatch of the IES. Subsequently, use the second-order cone programming method to solve the model and obtain the optimal operational status of the IES.

Step (4): Apply multiple calculations to obtain the experimental design of the state variables. Subsequently, establish the surrogate model

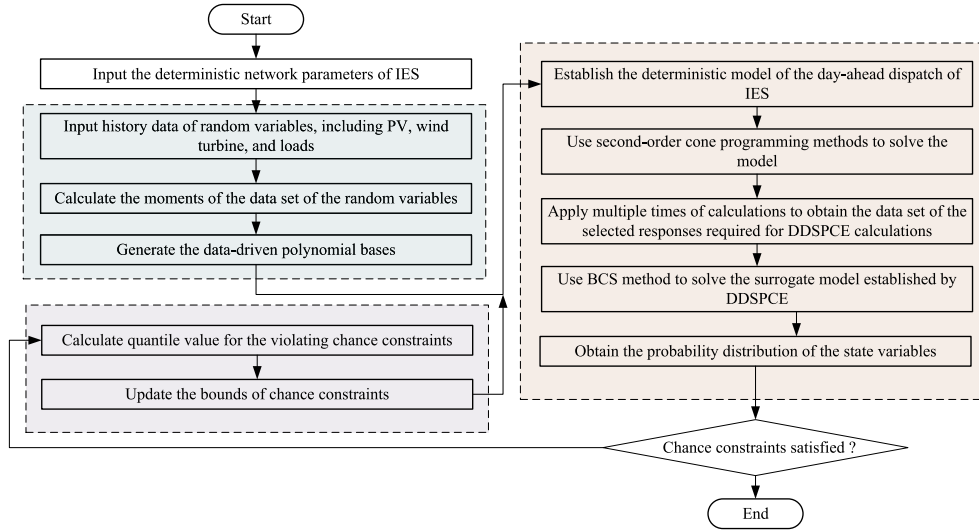


Fig. 6. Flowchart of the proposed chance-constrained optimal dispatch method based on DDSPE.

Table 1

Configuration of the energy stations.

Energy station	Configuration	Electricity node	Heat node	Gas node
ES1	CHP1	108	31	40
ES2	CHP2 (as heat slack node)	121	1	47
ES3	Electric boiler	91	32	–
ES4	Gas turbine	76	–	41

of these variables using the data-driven polynomial bases obtained in Step 2.

Step (5): Use the Bayesian compressive sensing method to solve the surrogate model. Subsequently, obtain the probabilistic distribution of the state variables.

Step (6): Use the probabilistic distribution to verify whether the chance constraints are satisfied. Otherwise, return to Step (3) and calculate the quantile value of the violating constraints to update the bounds in the deterministic model.

Results and discussions

Case study

A test IES case was used to verify the feasibility and advantages of the proposed method, in which the modified IEEE 123-bus electric network [45], 48-node gas network [46] and 32-node heat network [34]. The three subsystems are connected through four energy stations, the configurations of the energy stations are listed in Table 1. The electric, heat, and gas loads and output power of the PV and wind turbines were assumed to be random variables. The electricity and gas prices are shown in Fig. B.1 in Appendix “Electric price and gas price”. The generating information of original data of uncertain variables are shown in Appendix “Generating information of original data of uncertain variables”.

Three scenarios were considered, as follows:

Scenario 1: The operation strategy is obtained through deterministic optimization, where the average values of the random input variables are used as the deterministic input.

Scenario 2: The operation strategy is obtained through stochastic optimization, in which the optimization constraints must be satisfied in any random scenario. In other words, stochastic optimization is equivalent to CCO with a 100% confidence level.

Scenario 3: The operation strategy is obtained using the proposed CCO method with different confidence levels.

Calculation results

Based on Scenarios 1 to 3, the operation strategies of the test IES were calculated and analyzed.

Verification of PDFs obtained using DDSPE-BCS and MCS

Fig. 7 compares the PDFs of several example state variables of the integrated energy system (IES) in Scenario 1 and 3 using DDSPE-BCS and MCS. The comparison of the curve and histogram within the same scenario demonstrates that the PDFs obtained by the proposed DDSPE-BCS method closely align with the benchmark results using MCS, proving the accuracy of the proposed method. The comparison of Scenarios 1 and 3 demonstrates the PDFs of the state variables under different operation strategies. The operation strategy in Scenario 1 is obtained through deterministic optimization, while in Scenario 3 the strategy is obtained using the proposed CCO method with a confidence level of 10%. Taking the nodal voltage of electric bus 78 at 12:00 as an example and using the method presented in Fig. 5, the probability of violating lower voltage limit (0.95 p.u.) in Scenario 1 is 18.9%, while in Scenario 3 is reduced to 10%. The comparison result indicates that the proposed CCO method is effective to restrict the violation probability according to the confidence level. Similar conclusions can be drawn from other sub-figures.

Comparison of operation strategies

Fig. 8 shows the operation strategies of the heat network for 24 h in Scenarios 1 and 3. Considering the electric price shown in Fig. 2, the electric boiler was operated only in the low-electric-price interval between 1:00–8:00 and 13:00–19:00 daily. Thus, the heat supply must be provided by other equipment for the remainder of the day. HS operates in an opposite process to an electric boiler, where heat power is stored when the electric boiler is operating and released at other times. CHP2 produces heat based on its output electric power, which is determined by the operational status of the electric network. CHP1 is the balance node of the heat system, and the remaining heat load is satisfied by CHP1.

Fig. 9 shows the operation strategies of the electric network over 24 h. The distribution generation reached a consumption rate of 100%. The operational strategy for the electric boiler is described above. Because 1:00–8:00 and 13:00–19:00 are the intervals where the peak electric load and price occur, the electric storage must be charged in the low-electric-price interval.

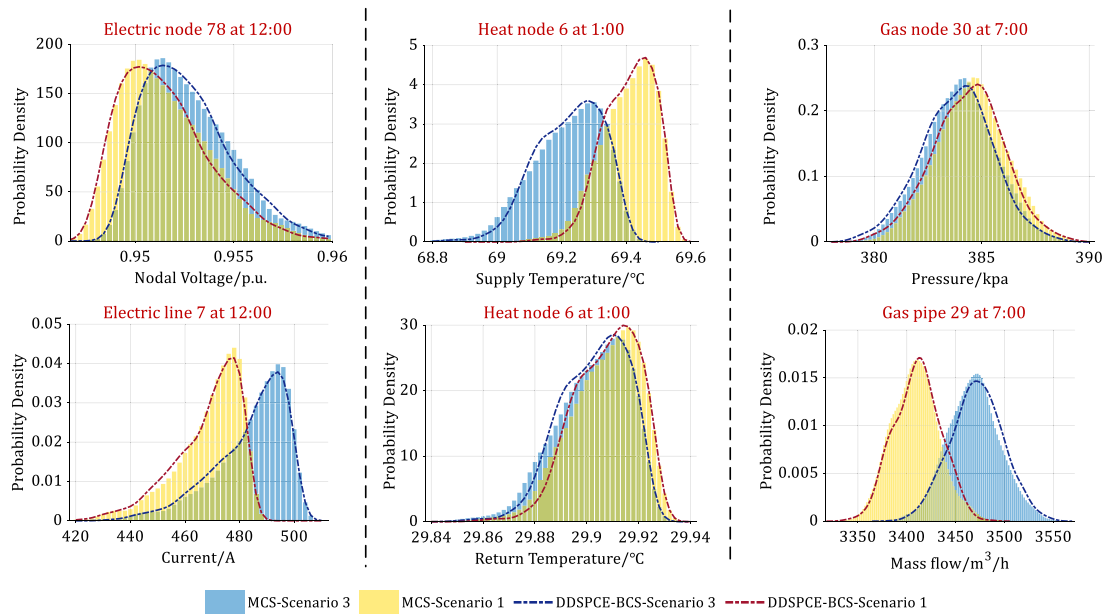


Fig. 7. Comparison of PDFs of state variables in Scenarios 1 and 3 using MCS and DDSPCE-BCS.

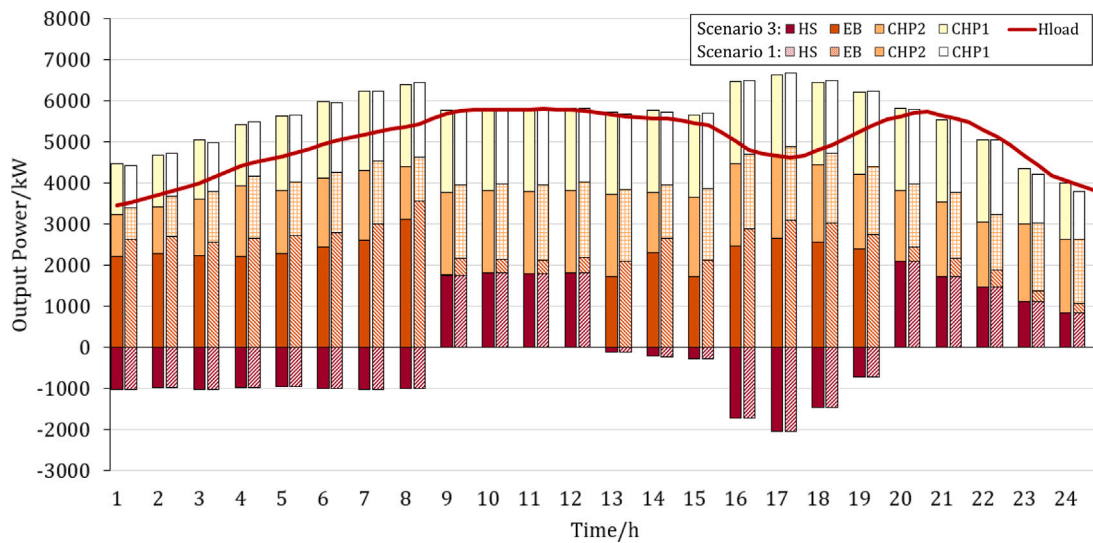


Fig. 8. Operation strategy of heat network.

Fig. 10 shows the 24-h input and output power and the SOC of the storage devices. As shown, the operation of thermal energy storage is affected significantly by the electricity price and electric boilers. Moreover, the HS must be completely discharged at 24:00 daily owing to the model constraints.

Fig. 11 shows the violation risk of the nodal voltages, supply temperature, and gas pressure for Scenarios 1 and 3, where red and blue denote the violation probabilities of the upper and lower bounds, respectively. As shown in Fig. 11(a), the operation strategy of Scenario 1 indicates a high risk of violating the lower bound around electric buses 77–97 and gas nodes 40–47, which is mainly caused by the overuse of CHP2 and EB. Scenario 3, which is a chance-constrained operation strategy, effectively restricts the violation probability by 10% in the total IES by reducing the output powers of CHP2 and EB, increasing the output powers of CHP1 and GB, and adjusting the compression rate of the compressor.

Comparison of objective function under different confidence levels

The objective functions of Scenarios 2 and 3, which are also the operation costs of the IES, were compared to validate the economic advantage of the proposed method. To further consider the effects of different confidence levels, Scenario 3 under confidence levels of 95%, 90%, 80%, 70%, and 60% were selected for comparison with Scenario 2, which is denoted as Scenario 3A–3E. The optimal dispatch problems under these confidence levels were solved, and the calculation results for the objective function are listed in Table 2.

Note that stochastic optimization method aims at 100% safe operation. However, the obtained strategy is overly conservative for the practical operation of the system. To avoid extreme scenarios with low probabilities, the operation strategy derived from stochastic optimization has the highest operation cost among all methods. This primary drawback of stochastic optimization is its excessive focus on

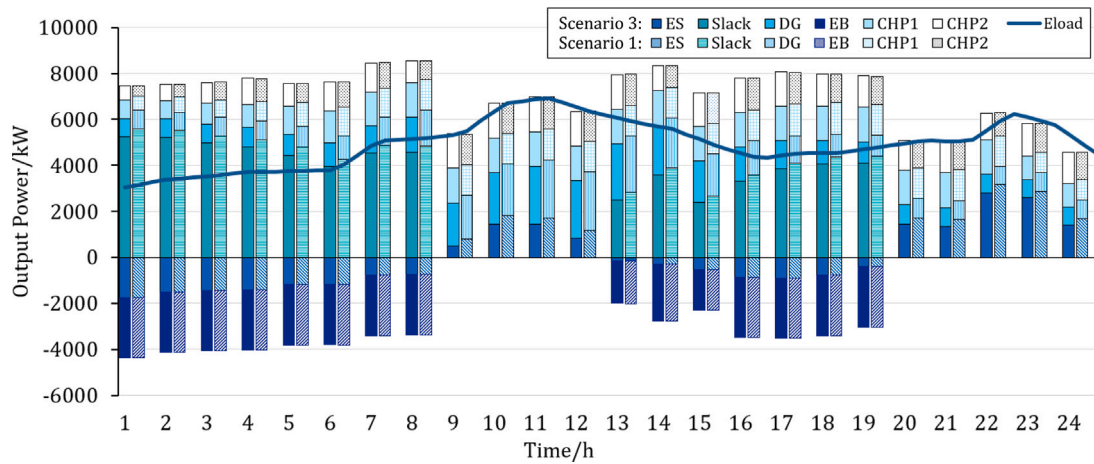


Fig. 9. Operation strategy of electric network.

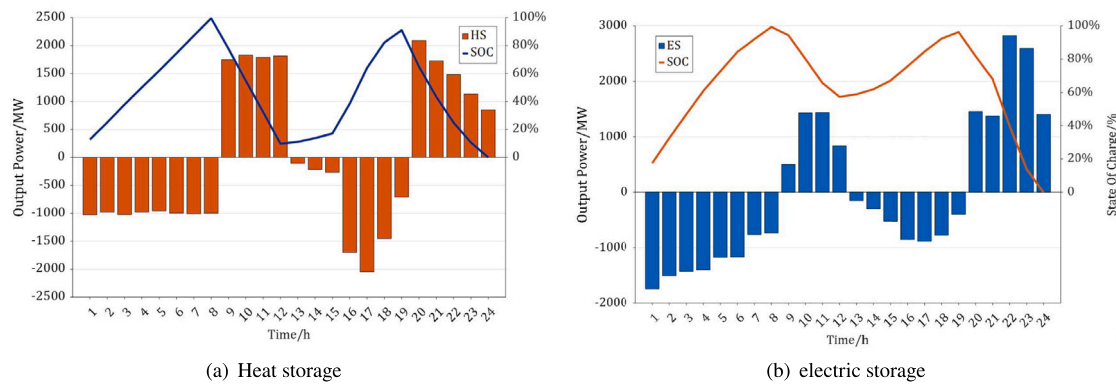


Fig. 10. Input/output power and SOC of storage devices.

Table 2

Comparison of the values of objective function under different confidence levels.

Scenarios	Confidence level	Objective function/USD	Reduction ratio
1	–	18 042.73	–8.41%
2	100% (equivalent)	19 486.62	–
3A	95%	19 218.60	–1.38%
3B	90%	18 630.07	–4.39%
3C	80%	18 505.73	–5.02%
3D	70%	18 476.71	–5.19%
3E	60%	18 356.52	–5.80%

the safety of the operation strategy, often at the expense of economic performance.

As shown in Table 2, comparing Scenarios 3A to 3E with Scenario 2, the operation cost of the system decreases with lower confidence level. Scenario 3E resulted in the lowest cost but the highest probability of voltage/flow/pressure violations (a maximum of 40%). Moreover, as the confidence level decreased to a certain point, the operation cost did not decrease significantly because the state variables of some nodes and pipelines can no longer be adjusted to exceed the confidence level.

Scenario 1, i.e., the deterministic optimization method, indicates the lowest operating cost because it ignores the influence of uncertainties. Deterministic optimization only focuses on the economic efficiency of system operation but cannot deal with the potential risks that the system may face. Therefore, compared to stochastic optimization or chance-constrained optimization, the operational strategies derived

from deterministic optimization expose the system to significant operational risks. As can be seen in Fig. 11(b), the operation strategy of Scenario 1 has a high risk of violation, which may lead to operational failures or accidents.

In summary, the advantages of chance-constrained optimization compared to other methods are as follows:

(1) Chance-constrained optimization can effectively deal with the uncertainties in IES, restrict the violation risk to a predefined confidence level, and enhance the ability of the system to deal with unexpected events and the impact of uncertainties. Compared with deterministic optimization, the maximum violation risk decreased from 36% to 10%.

(2) Chance-constrained optimization can provide more flexibility in balancing operational risks and benefits, avoiding overly conservative operational strategies for preventing low probability extreme scenarios. Compared with stochastic optimization, the chance-constrained optimization under a confidence level of 90% can reduce the operation cost by 4.39%.

Advantage of improved iterative procedure

Next, the performance of the proposed improved iterative algorithm was verified. Fig. 12 shows the number of iteration steps and convergence accuracy of the original and improved iterative algorithms when computing the test IES case. The improved iterative algorithm

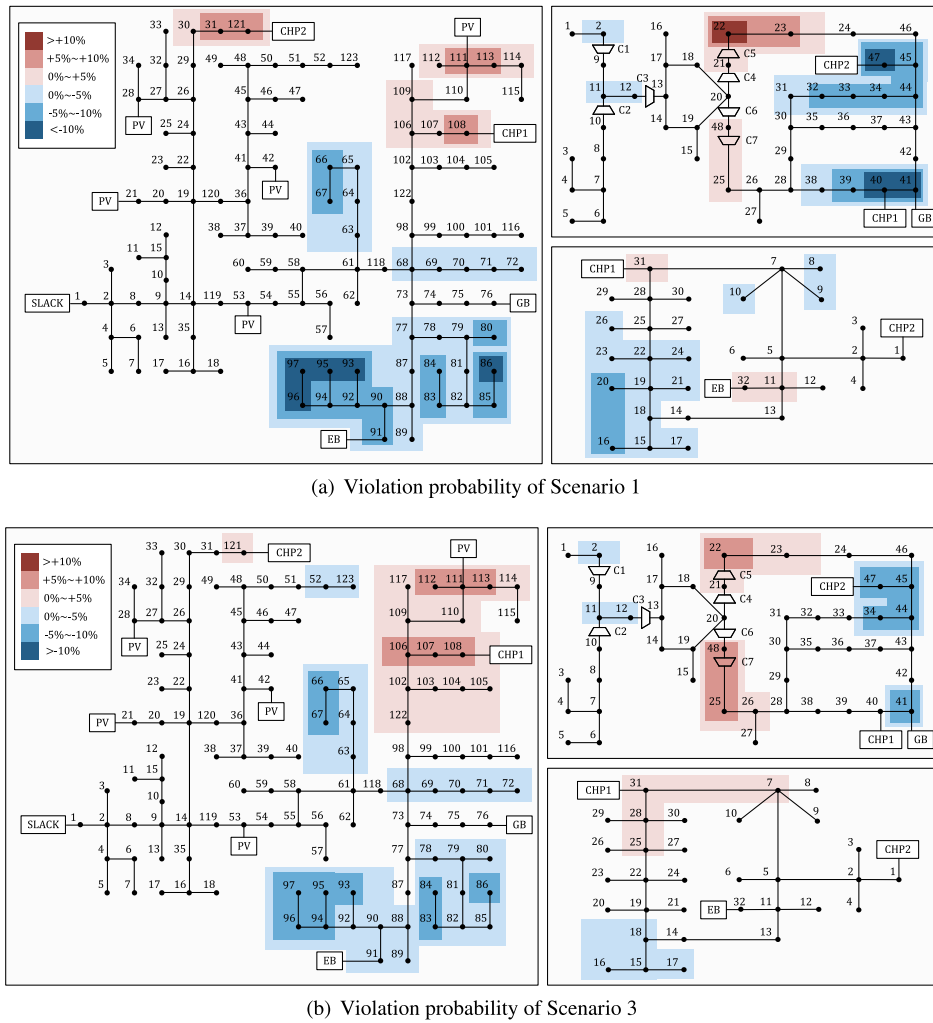


Fig. 11. Comparison of violation probabilities of IES in Scenarios 1 and 3.

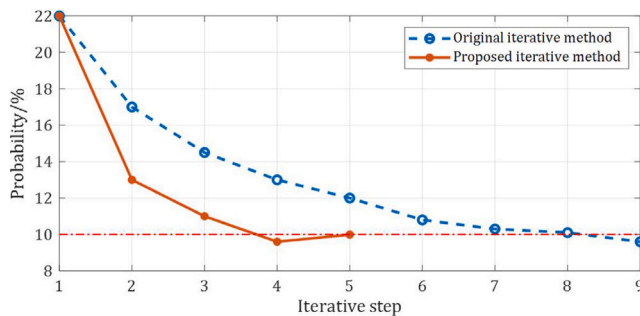


Fig. 12. Comparison of the convergence of the proposed and original iterative method.

required only five steps to converge, whereas the original iterative algorithm required nine steps. Moreover, the improved iterative algorithm converged at the value of 9.92%, whereas the original algorithm converged at 9.6%, which implies that the improved algorithm can achieve better convergence on the predefined confidence level, thus reducing the conservation of the obtained optimization strategy. For further guarantee of the convergence of iterative algorithm structure, we refer to [47] for detailed theory analysis and large-scale test case.

Conclusion

A DDSPCE method for the chance-constrained optimal dispatch of an IES was proposed herein. By adopting DDSPCE, numerous dependent samples were efficiently evaluated to estimate the violation probability for each chance constraint, and the accuracy was verified via comparison with MCS. Moreover, the proposed improved iterative structure for the verification of chance constraints effectively reduced the number of iterative steps and increased the calculation speed. These contributions allowed us to obtain a fast and accurate solution for the chance-constrained optimal dispatch. Finally, the proposed method was applied to a test case for an IES, and the obtained optimization strategy was evaluated and compared using deterministic and stochastic optimization strategies. The results showed that, compared with the deterministic strategy, the proposed method effectively controlled the violation probability to the preset confidence level, thus ensuring the safe operation of the IES. Compared with the conservative stochastic optimization method, the proposed method reduced the operational cost by 4.39% at a confidence level of 90%. Selecting a lower confidence level allows the operational cost to be further reduced while ensuring operational safety.

However, this study only considered the steady-state model of the IES; in actual applications, the transmission delay of gas and heat

networks should be not negligible. Thus, a semi-dynamic or dynamic model of the energy subsystems can be considered in future studies. Using these models, the characteristics of a large time constant system can be considered, which renders the calculation results more realistic and promising for further study.

CRedit authorship contribution statement

Bo Dong: Data curation, Writing – original draft. **Peng Li:** Conceptualization, Methodology. **Hao Yu:** Supervision, Writing – review & editing. **Haoran Ji:** Investigation, Software. **Guanyu Song:** Software. **Juan Li:** Formal analysis. **Jinli Zhao:** Validation. **Chengshan Wang:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Operation constraints of DG, ES, and HS

The operational constraints of DGs are expressed as follows:

$$-P_{t,i}^{DG} \tan \theta_i^{DG} \leq Q_{t,i}^{DG} \leq P_{t,i}^{DG} \tan \theta_i^{DG} \quad (A.1)$$

$$\sqrt{(P_{t,i}^{DG})^2 + (Q_{t,i}^{DG})^2} \leq S_i^{DG} \quad (A.2)$$

where $P_{t,i}^{DG}$ and $Q_{t,i}^{DG}$ denotes the active and reactive power of the i th DG at time t , respectively; θ_i^{DG} denotes the power factor angle of the i th DG; and S_i^{DG} denotes the capacity of the i th DG.

Constraints (A.1) and (A.2) represent the reactive power and capacity constraints, respectively.

The operational constraints of ES are expressed as follows:

$$\sqrt{(P_{t,i}^{ES})^2 + (Q_{t,i}^{ES})^2} \leq S_i^{ES} \quad (A.3)$$

$$P_{t,i}^{ES,loss} = \eta_i^{ES} \sqrt{(P_{t,i}^{ES})^2 + (Q_{t,i}^{ES})^2} \quad (A.4)$$

$$Q_{i,min}^{ES} \leq Q_{t,i}^{ES} \leq Q_{i,max}^{ES} \quad (A.5)$$

$$E_{t+1,i}^{ES} = E_{t,i}^{ES} - (P_{t,i}^{ES} + P_{t,i}^{ES,loss}) \Delta t \quad (A.6)$$

$$E_{i,min}^{ES} \leq E_{t,i}^{ES} \leq E_{i,max}^{ES} \quad (A.7)$$

$$E_{i,1}^{ES} = E_{i,24}^{ES} \quad (A.8)$$

where $P_{t,i}^{ES}$ and $Q_{t,i}^{ES}$ denote the active and reactive powers of the i th ES at period t , respectively; S_i^{ES} denotes the capacity of the i th electric storage at period t ; $P_{t,i}^{ES,loss}$ denotes the loss power of the i th ES at period t ; $Q_{i,max}^{ES}$ and $Q_{i,min}^{ES}$ denote the maximum and minimum limit of the reactive power of the i th ES, respectively; $E_{t,i}^{ES}$ denotes the energy storage of the i th ES at period t ; and $E_{i,max}^{ES}$ and $E_{i,min}^{ES}$ denote

the maximum and minimum limits of energy storage of the i th ES, respectively.

Constraint (A.3) represents the capacity limit of ES. The power loss of the ES is considered in constraint (A.4), and the reactive power limit of the ES is shown in constraint (A.5). Constraint (A.6) shows the calculation of the stored energy of the ES in period $t + 1$. Constraint (A.7) indicates the ES energy storage limit. Based on constraint (A.8), the energy storage at 1:00 and 24:00 should be equal to ensure the continuity and sustainability of the operation.

The operational constraints of HS are expressed as follows:

$$E_{i,t+1}^{HS} = (1 - \epsilon^{HS}) S_{i,t}^{HS} + P_{i,t}^{HS,C} \eta_{i,C} \Delta t - P_{i,t}^{HS,D} \eta_{i,D} \Delta t \quad (A.9)$$

$$0 \leq P_{i,t}^{HS,C} \leq P_{i,max}^{HS,C} \quad (A.10)$$

$$0 \leq P_{i,t}^{HS,D} \leq P_{i,max}^{HS,D} \quad (A.11)$$

$$E_{i,min}^{HS} \leq E_{i,t}^{HS} \leq E_{i,max}^{HS} \quad (A.12)$$

$$E_{i,1}^{HS} = E_{i,24}^{HS} \quad (A.13)$$

where $E_{i,t}^{HS}$ denotes the energy storage of the i th HS heat at period t ; ϵ^{HS} denotes the self-discharging rate of HS; $P_{i,t}^{HS,C}$ and $P_{i,t}^{HS,D}$ denote the charging and discharging powers of the i th HS heat at period t , respectively; $P_{i,max}^{HS,C}$ and $P_{i,max}^{HS,D}$ denote the maximum charging and discharging powers of the i th HS, respectively; $\eta_{i,C}$ and $\eta_{i,D}$ denote the efficiency of charge and discharge, respectively; and $E_{i,min}^{HS}$ and $E_{i,max}^{HS}$ denote the minimum and maximum state of charge of the i th HS, respectively.

Appendix B. Electric price and gas price

See Fig. B.1.

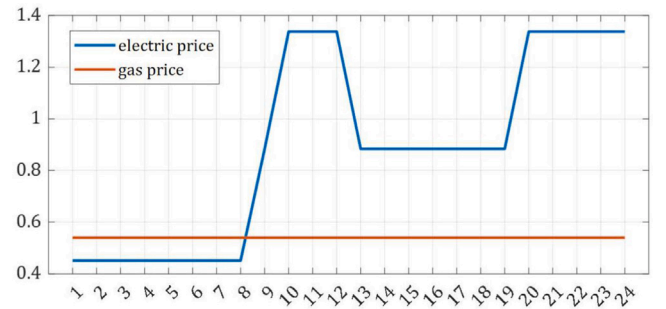


Fig. B.1. Electric price and gas price.

Appendix C. Generating information of original data of uncertain variables

See Table C.1.

Table C.1

Generating information of original data of uncertain variables.

Uncertain variables	Distributions	Parameters
Electric/heat/gas load	Beta distribution & Gaussian distribution	$\alpha = 0.8, \beta = 2.5 \& \mu = 0, \sigma^2 = 0.5$
Photovoltaic	Beta distribution & Uniform distribution	$\alpha = 0.8, \beta = 2.5 \& a = 0.8, b = 1.2$
Wind turbine	Weibull distribution & Uniform distribution	$k = 2, c = 0.5 \& a = 0.8, b = 1.2$

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